

Roy Kim

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TECHNICAL SKILLS

Languages: Python, Java, Go, SQL, JavaScript, C, Bash, Perl, HTML/CSS

Technologies: Flask, FastAPI, React, NumPy, Pandas, PostgreSQL, MySQL, Kafka, Git, Linux, Jenkins, Docker

EXPERIENCE

Backend Software Developer – Compute and Storage Engineering

June 2023 – Present

Morgan Stanley

Montreal, QC

- Developed the software distribution process for trading infrastructure across 1,400+ DMZ and colocation servers
- Modernized a legacy CLI tool into a full-stack web application featuring a real-time dashboard for client update status across global packager distribution queues
- Parallelized the packaging pipeline with multiprocessing, cutting distribution time by 92%
- Led a cross-environment reliability overhaul, resolving dev/test path-resolution failures across distributed and local environments over a month-long effort, establishing the pattern later adopted by other tooling
- Consolidated five regionally mounted NFS shares into a unified cloud-based Artifactory repository, building an automated publishing pipeline for generic and container image artifacts
- Increased test coverage from 30% to over 70% by expanding the automated suite to 260+ test cases, including fixing a silently broken test teardown that had left cleanup not running between runs
- Rolled out Kerberos-based authentication and certificate-based authorization across service accounts and human users, replacing a legacy authentication path

IT Developer – Security Access Management Infrastructure

May 2022 – June 2023

Canada Revenue Agency

Ottawa, ON

- Designed and implemented scalable, fault-tolerant infrastructure for API Gateway supporting 45 external clients
- Migrated all legacy Solaris and RHEL6 systems to RHEL7+, ensuring continued Red Hat support
- Onboarded 2 new interns and full-time employees
- Supported bi-weekly on-call rotations for infrastructure-related outages

IT Developer Intern – Security Access Management Operations

May 2020 – May 2022

Canada Revenue Agency

Ottawa, ON

- Optimized the server daily health check by 90% by removing redundant and inconsistent data
- Reduced manual effort and increased accuracy of bulk Active Directory modification requests by building an automated process that parses request data and generates LDIF files
- Reduced outage rectification time by streamlining recovery procedures for authentication data outages
- Built a Git repository of 87 custom scripts and documentation used across 40+ servers

PROJECTS

Mechmarket Inventory

Python, MySQL, FastAPI, Requests

- Developed a bot leveraging the Reddit API and Python Requests to parse over 700,000 posts from r/mechmarket
- Designed a product dashboard to view fluctuating pricing of artisan products

Classroom Attendance

C++, Linux, libbluetooth, MySQL

- Built a C++ attendance tool to help professors document individual student participation
- Leveraged a Raspberry Pi's Bluetooth capabilities via Linux libbluetooth-dev and BlueZ to scan and record student attendance
- Compiled student Bluetooth MAC addresses into a MySQL database to generate individual student reports

EDUCATION

Western University – B.S. in Computer Science

2017 – 2022